PORTFOLIO ARTIFACT · AIML-501 MODEL DEVELOPMENT · INDIANA WESLEYAN UNIVERSITY

Machine Learning Algorithms: A Visual Framework

Twelve core algorithms organized by how they learn, where they are used, and what they are good for, from spreadsheet predictions to generative AI.

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HOW TO READ THIS FRAMEWORK

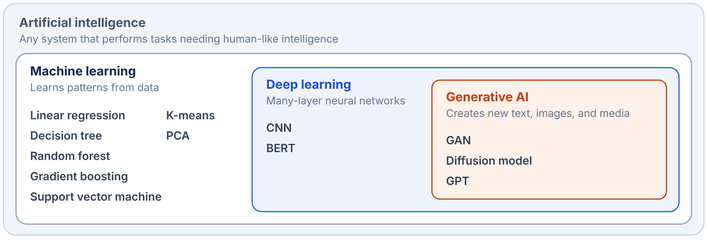
Supervised — **Learns from labeled examples** (inputs paired with correct answers) to predict a label or a number.

Unsupervised — **Finds structure in unlabeled data.** Self-supervised models are a form of this: they hide part of the data and learn to predict it.

APPLICATION DOMAINS

Tabular data · Computer vision · NLP · Generative AI

WHERE EACH ALGORITHM SITS: AI ⊃ ML ⊃ DEEP LEARNING ⊃ GENERATIVE AI



01 Supervised learning

These algorithms need labeled training data. They power most predictive analytics: forecasting a number (regression) or assigning a category (classification).

Linear Regression
Predicts a number

Supervised · Tabular data

HOW IT WORKS

Fits the straight line (or plane) that keeps the gap between predicted and actual values as small as possible, then reads predictions off that line.

USE CASES

- Forecasting monthly sales from ad spend
- Estimating house prices from size and location

Decision Tree
Rule-based yes/no splits

Supervised · Tabular data

HOW IT WORKS

Asks a series of yes/no questions about the features, choosing each question to make the groups purer, until every leaf holds one answer.

USE CASES

- Transparent loan-approval rules
- Sorting fruit by shape, color, and texture

Random Forest
Many trees that vote

Supervised · Tabular data

HOW IT WORKS

Trains hundreds of decision trees on random samples of rows and features, then takes a majority vote, so individual trees' mistakes cancel out.

USE CASES

- Credit card fraud detection
- Predicting hospital readmission risk

Gradient Boosting
XGBoost, LightGBM

Supervised · Tabular data

HOW IT WORKS

Builds small trees one after another; each new tree focuses on the errors the earlier trees left behind. Often the most accurate choice for spreadsheet-style data.

USE CASES

- Credit scoring
- Predicting which customers will cancel

Support Vector Machine
Widest-margin boundary

Supervised · Tabular data · NLP

HOW IT WORKS

Finds the boundary that separates two classes with the widest possible margin; "kernels" let it draw curved boundaries when a straight line will not work.

USE CASES

- Email spam filtering
- Classifying tumors as benign or malignant

Convolutional Neural Network
CNN

Supervised · Computer vision

HOW IT WORKS

Slides small learned filters across an image to detect edges, then shapes, then whole objects, building understanding layer by layer.

USE CASES

- Spotting pneumonia in chest X-rays
- Detecting pedestrians for self-driving cars

02 Unsupervised and self-supervised learning

These algorithms learn from raw, unlabeled data. Classic methods group or simplify data; self-supervised and generative models learn the structure of language and images well enough to create new content.

K-means Clustering
Groups similar items

Unsupervised · Tabular data

HOW IT WORKS

Places k centers, assigns each point to its nearest center, moves each center to the average of its points, and repeats until the groups stop changing.

USE CASES

- Customer segmentation for marketing
- Grouping similar products or documents

Principal Component Analysis
PCA

Unsupervised · Tabular data · Computer vision

HOW IT WORKS

Rotates the data onto new axes that capture the most variation, so dozens of related features can be summarized by a few components.

USE CASES

- Reducing hundreds of sensor readings to a few signals
- Plotting complex data in two dimensions

BERT
Transformer encoder

Self-supervised, then fine-tuned · NLP

HOW IT WORKS

Learns language by predicting hidden words from the context on both sides, then is fine-tuned with a small labeled dataset for a specific task.

USE CASES

- Sentiment analysis of product reviews
- Understanding search queries

Generative Adversarial Network
GAN

Unsupervised · Generative AI · Computer vision

HOW IT WORKS

A generator creates fake samples while a discriminator tries to catch them. Competing against each other, the generator learns to produce realistic output.

USE CASES

- Synthetic medical images to enlarge small datasets
- Photo upscaling and restoration

Diffusion Model
e.g., Stable Diffusion

Unsupervised · Generative AI · Computer vision

HOW IT WORKS

Learns to remove noise from images step by step. To create, it starts from pure noise and denoises toward an image that matches a text prompt.

USE CASES

- Text-to-image design concepts
- Editing or filling in parts of a photo

GPT
Transformer decoder (large language model)

Self-supervised · NLP · Generative AI

HOW IT WORKS

Predicts the next word from all the words before it using "attention." Trained on huge text collections, then refined with human feedback.

USE CASES

- Chat assistants and writing support
- Summarizing reports and generating code

DOMAIN COVERAGE AT A GLANCE

ALGORITHM	TYPE	TABULAR	VISION	NLP	GEN AI
Linear regression	Supervised	●			
Decision tree	Supervised	●			
Random forest	Supervised	●			
Gradient boosting	Supervised	●			
SVM	Supervised	●	○	○	
CNN	Supervised		●		○
K-means	Unsupervised	●	○		
PCA	Unsupervised	●	○		
BERT	Self-supervised			●	
GAN	Unsupervised		●		●
Diffusion model	Unsupervised		●		●
GPT	Self-supervised			●	●

● Primary domain ○ Also used

WHICH ALGORITHM FITS THE PROBLEM?

What do you need the model to do?

Predict a known label or number

Spreadsheet-style data
Gradient boosting or random forest; linear regression or a decision tree when explanations matter most

Images
CNN

Text
BERT (or SVM for small, simple text tasks)

Find hidden patterns without labels

Group similar records
K-means clustering

Simplify many features
PCA

Then predict
Feed the clusters or components into a supervised model

Create new content

Text, chat, code
GPT-style language model

Images from a prompt
Diffusion model

Realistic synthetic samples
GAN

Start from the goal and the data type, then weigh accuracy against how easily people need to understand the model's decisions.

Key sources: Breiman (2001) Random forests · Chen & Guestrin (2016) XGBoost · Cortes & Vapnik (1995) Support-vector networks · Devlin et al. (2019) BERT · Goodfellow et al. (2014) Generative adversarial nets · Ho et al. (2020) Denoising diffusion probabilistic models · James et al. (2021) An introduction to statistical learning · LeCun et al. (1998) Gradient-based learning applied to document recognition · MacQueen (1967) K-means · Radford et al. (2018) Generative pre-training · Vaswani et al. (2017) Attention is all you need. Full APA references are listed on the portfolio page.